

EXPLORING CYTOMEGALOVIRUS



FINAL PROJECT REPORT SUBMITTED
IN FULFILMENT OF THE REQUIREMENTS FOR COURSE STAT 482
CATEGORICAL DATA ANALYSIS
DEPARTMENT OF STATISTICS OF MIDDLE EAST TECHNICAL
UNIVERSITY

BY

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1. Did you use ChatGPT or a similar LLM program in any way in statistical or writing parts of this project?

I used Grammarly to see my spelling and grammar mistakes of the report and citation website for citing for references part .Other than that, I did not use any programs.

2. Did you get help from another person in analyzing the data or writing the report or making the dashboard?

NO.

ABSTRACT

Cytomegalovirus (CMV) is still a major public health risk because of its frequency and possible side effects. In order to look into the parameters connected to CMV infection, this study uses logistic regression modeling and exploratory data analysis (EDA) methodologies. By employing a dataset that includes clinical, laboratory, and demographic data from patients, my research provides new perspectives on epidemiology of CMV. Important trends in the data are found by EDA, providing information about distribution and connections between the variables. Logistic regression analysis is then used to measure the relationship between different predictors and the probability of contracting CMV infection. The findings provide important new information for comprehending and treating CMV infections by highlighting important factors.

1. Introduction

In this work, I use both EDA and Logistic Regression techniques to provide a thorough investigation of CMV infection. By utilizing a dataset that includes clinical, laboratory, and demographic data from a group of patients, my investigation looks for clarifying the multiple factors that contribute to CMV infection. Through a methodical analysis of the interactions between different predictors and CMV status, I aimed to identify important variables and clarify their roles in CMV

1.1 Data description

This dataset includes 64 consecutive patients who received an allogeneic hematopoietic stem cell transplant with lower intensity conditioning from a matched sibling donor and T-cell depletion. (Additional information follows the variable data).

1. **ID:** Patient ID, a numeric identifier ranging from 1 to 64.
2. **age:** Recipient age at transplant, a numeric variable ranging from 29 to 67.
3. **sex:** Recipient sex, a binary variable where 0 indicates male and 1 indicates female.
4. **race:** Recipient race, a binary variable where 0 indicates white and 1 indicates African-American.
5. **diagnosis:** Category of cancer diagnosis, a character variable with 13 levels.

6. **diagnosis.type**: Cancer diagnosis type, ranging from 0 (myeloid) to 1 (lymphoid).
7. **time.to.transplant**: Time from cancer diagnosis to transplant (in months), a numeric variable ranging from 1.84 to 173.8.
8. **prior.radiation**: Prior radiation therapy, a binary variable where 0 indicates no prior radiation therapy and 1 indicates prior radiation therapy.
9. **prior.chemo**: Number of prior chemotherapy regimens, ranging from 0 to 8.
10. **prior.transplant**: Prior stem cell transplant, a binary variable represented numerically where 0 indicates no prior stem cell transplant and 1 indicates prior stem cell transplant.
11. **recipient.cmv**: Recipient cytomegalovirus seropositive status, a binary variable where 0 indicates negative and 1 indicates positive.
12. **donor.cmv**: Donor cytomegalovirus seropositive status, a binary variable 0 indicates negative and 1 indicates positive.
13. **donor.sex**: Donor sex, a binary variable where 0 indicates female and 1 indicates male.
14. **TNC.dose**: Total nucleated cell dose ($\times 10^8/\text{kg}$), a numeric variable ranging from 2.06 to 21.0.
15. **CD34.dose**: Total CD34+ (stem) cell dose ($\times 10^8/\text{kg}$), a numeric variable ranging from 2.04 to 12.5.
16. **CD3.dose**: Total CD3+ (T) cell dose ($\times 10^8/\text{kg}$), a numeric variable ranging from 1.08 to 8.2.
17. **CD8.dose**: Total CD8+ cell dose ($\times 10^8/\text{kg}$), a numeric variable ranging from 0.16 to 3.2.
18. **TBI.dose**: Total body irradiation dosage (centiGrays), a numeric variable ranging from 200.00 to 400.0.
19. **C1/C2**: HLA-Cw group, a binary variable where 0 indicates heterozygous and 1 indicates homozygous.
20. **aKIRs**: Number of donor activating killer immunoglobulin-like receptors, a numeric variable ranging from 1.00 to 6.0.
21. **cmv**: Cytomegalovirus reactivation posttransplant, a binary variable where 0 indicates No and 1 indicates Yes.
22. **time.to.cmv**: Time to cytomegalovirus reactivation (in months), a numeric variable ranging from 0.43 to 84.5.
23. **agvhd**: Acute level 2-4 graft versus host disease, a binary variable where 0 indicates No and 1 indicates Yes.
24. **time.to.agvhd**: Time to acute level 2-4 graft versus host disease (in months), a numeric variable ranging from 0.66 to 85.2.
25. **cgvhd**: Chronic graft versus host disease, a binary variable where 0 indicates No and 1 indicates Yes.
26. **time.to.cgvhd**: Time to chronic graft versus host disease (in months), a numeric variable ranging from 0.82 to 65.1.

1.2. Research Questions

- What is the association between variables ? Are they related to each other ?

- Which predictor variables contribute significantly to the logistic regression model for predicting CMV reactivation?
- How accurately can the logistic regression model predict CMV reactivation based on the available patient data?

1.3.Aim of the study

Creating a predictive model for cytomegalovirus (CMV) reactivation in recipients of stem cell transplants is the study's main objective. One common post-transplantation consequence that can cause serious health problems is CMV reactivation.

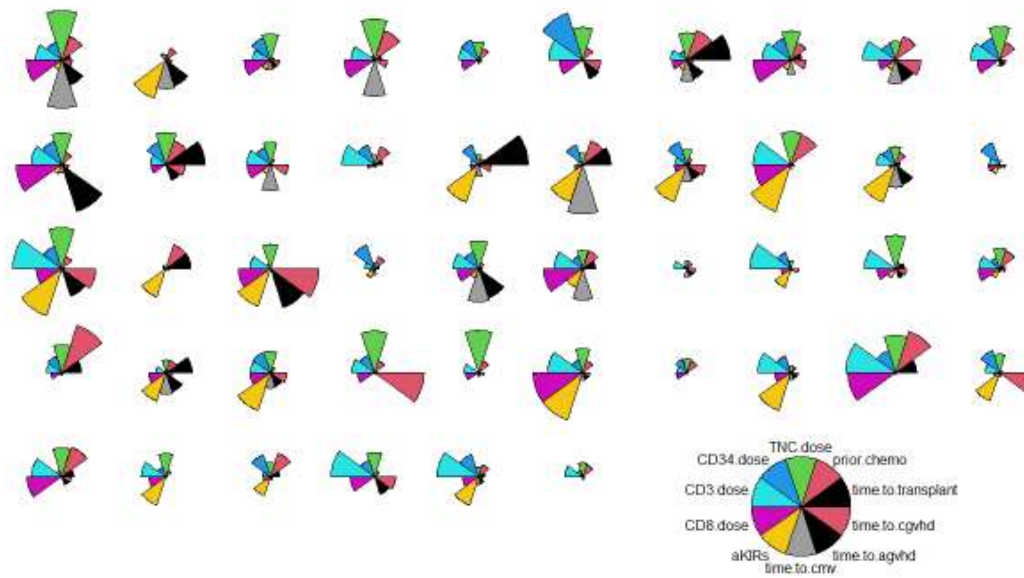
2. Methodology or Analysis

The cytomegalovirus (CMV) reactivation and stem cell transplantation patient data was analyzed using the R programming language in this study. Exploratory data analysis (EDA) was carried out in order to explore the features of the dataset, such as demographics. In order to handle missing values and encode categorical variables, data preparation techniques were used. Relationships between predictor variables and CMV reactivation status were found using statistical tests such chi-square tests and contingency tables. The glm() function in R was then used to create a logistic regression model that included appropriate predictor variables found by exploratory data analysis (EDA) and statistical studies. Performance of the model was assessed using criteria such as accuracy, sensitivity and specificity.

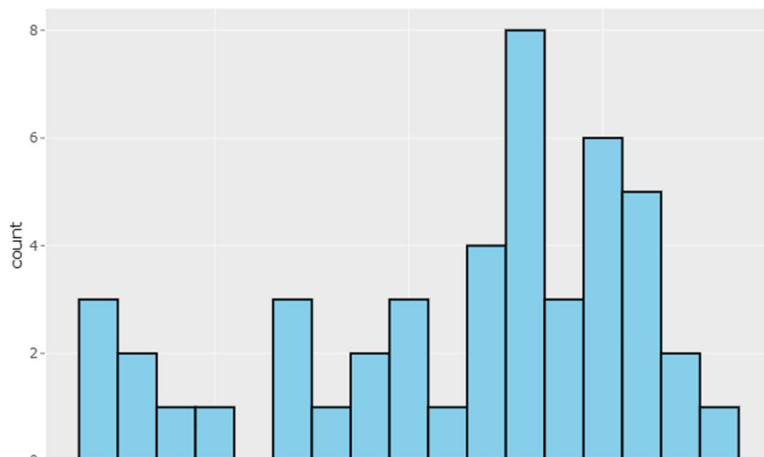
3.Results and Findings

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Stars plot



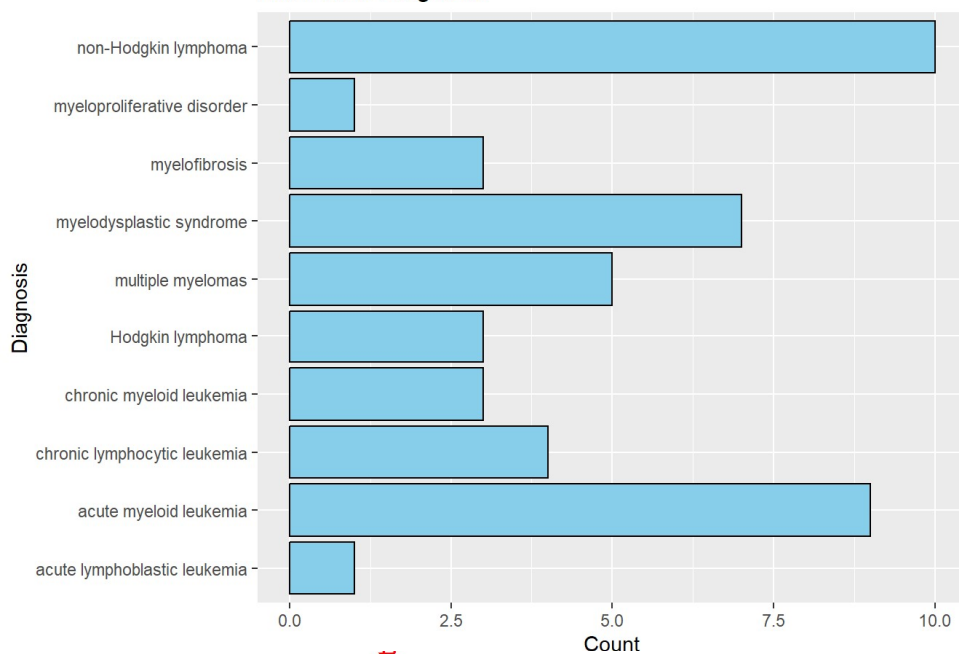
In this plot, I ordered the data with respect to increasing age. And The plot shows the numerical variables. From the plot, we can see that none of the variables are increasing or decreasing with age.



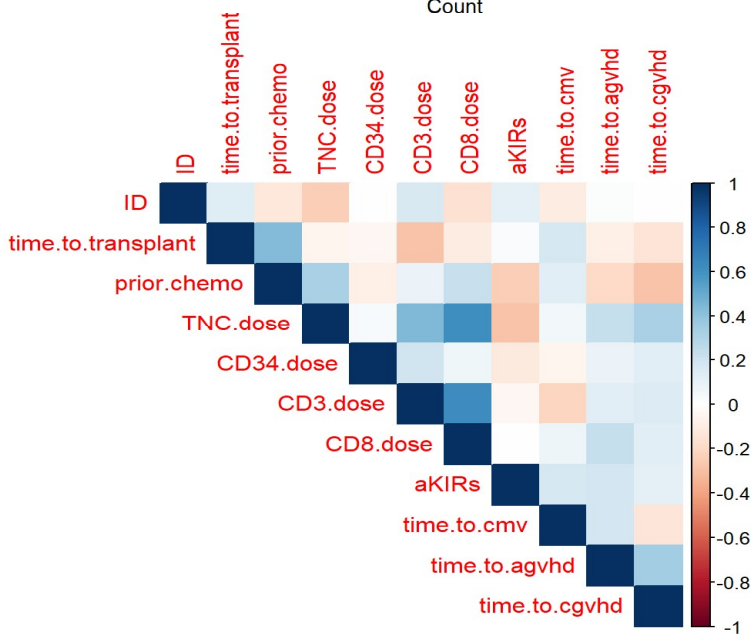
Bar Plot of Diagnosis

The majority of patients in the data set are between 50-60 years old.

The mean age of the patients is approximately 53.



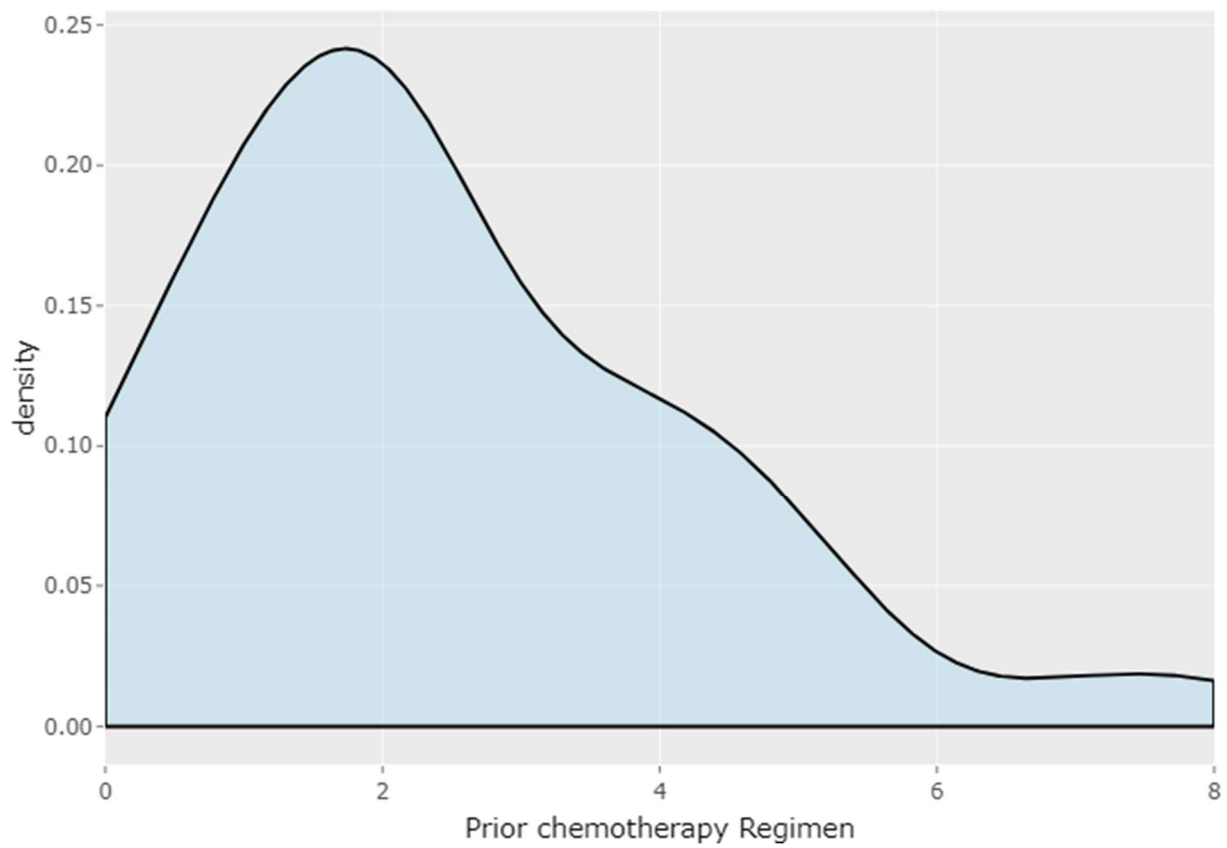
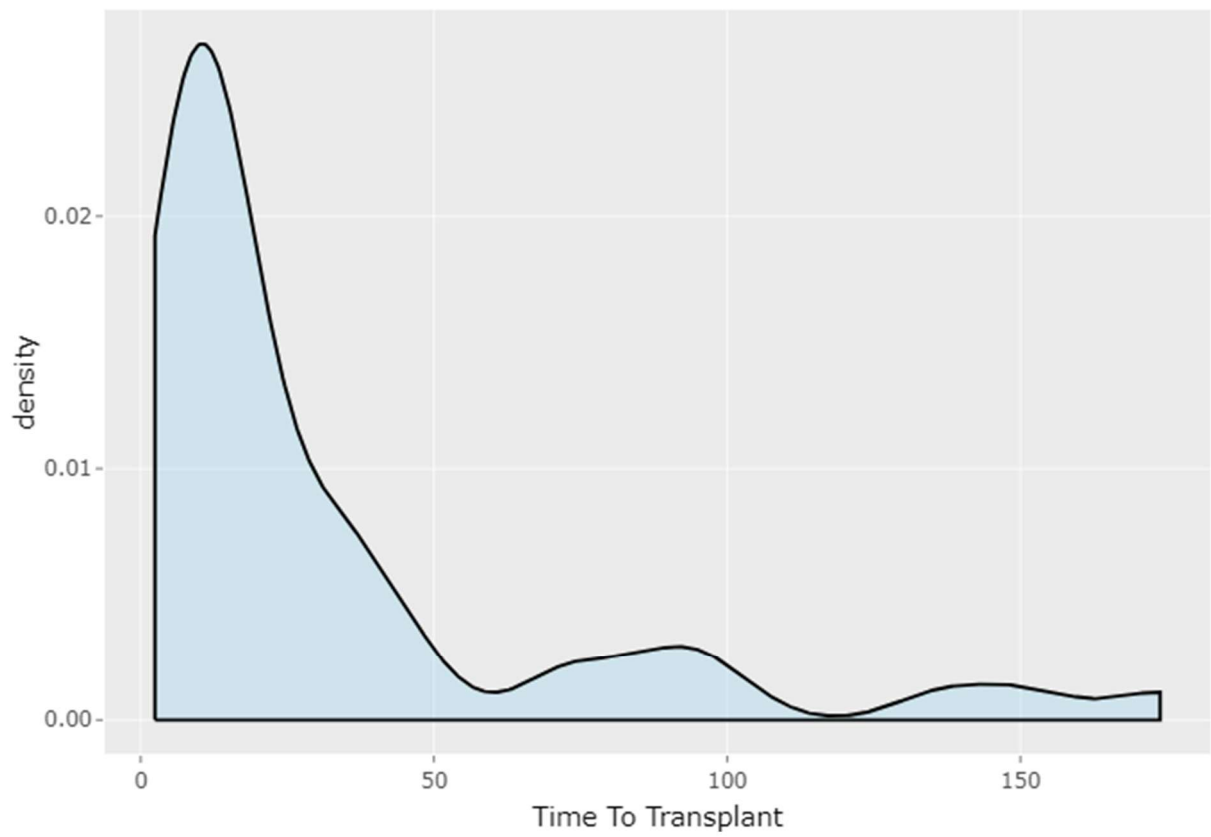
Here is the bar plot of diagnosis and their frequencies. Majority of people suffer from non-Hodgkin Lymphoma and Acute Myeloid Leukemia.

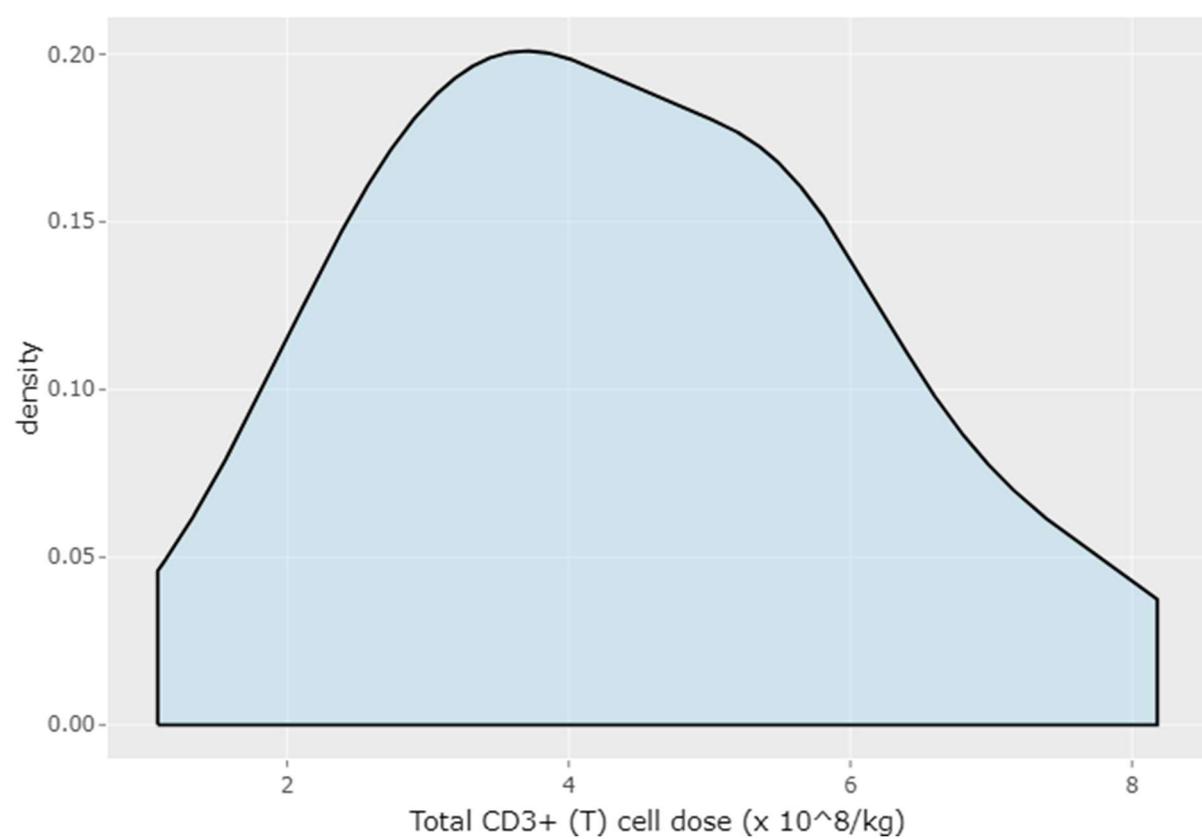
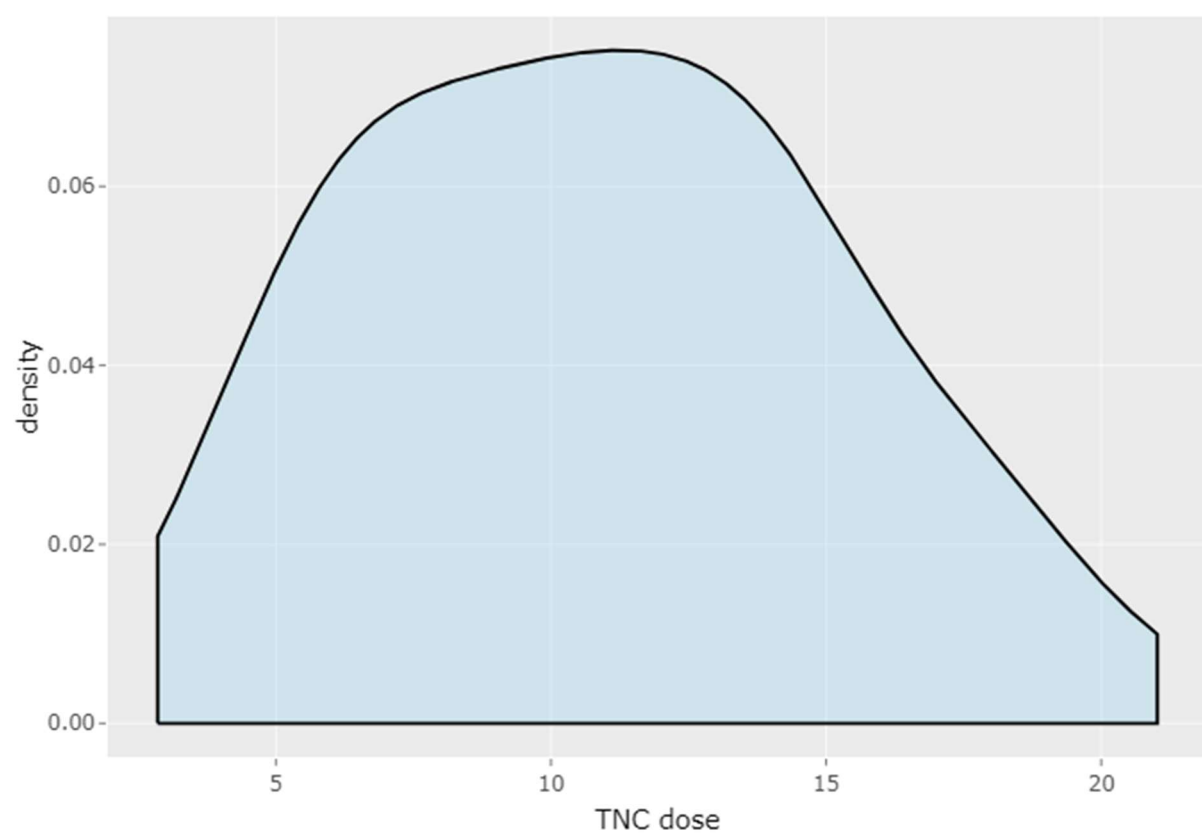


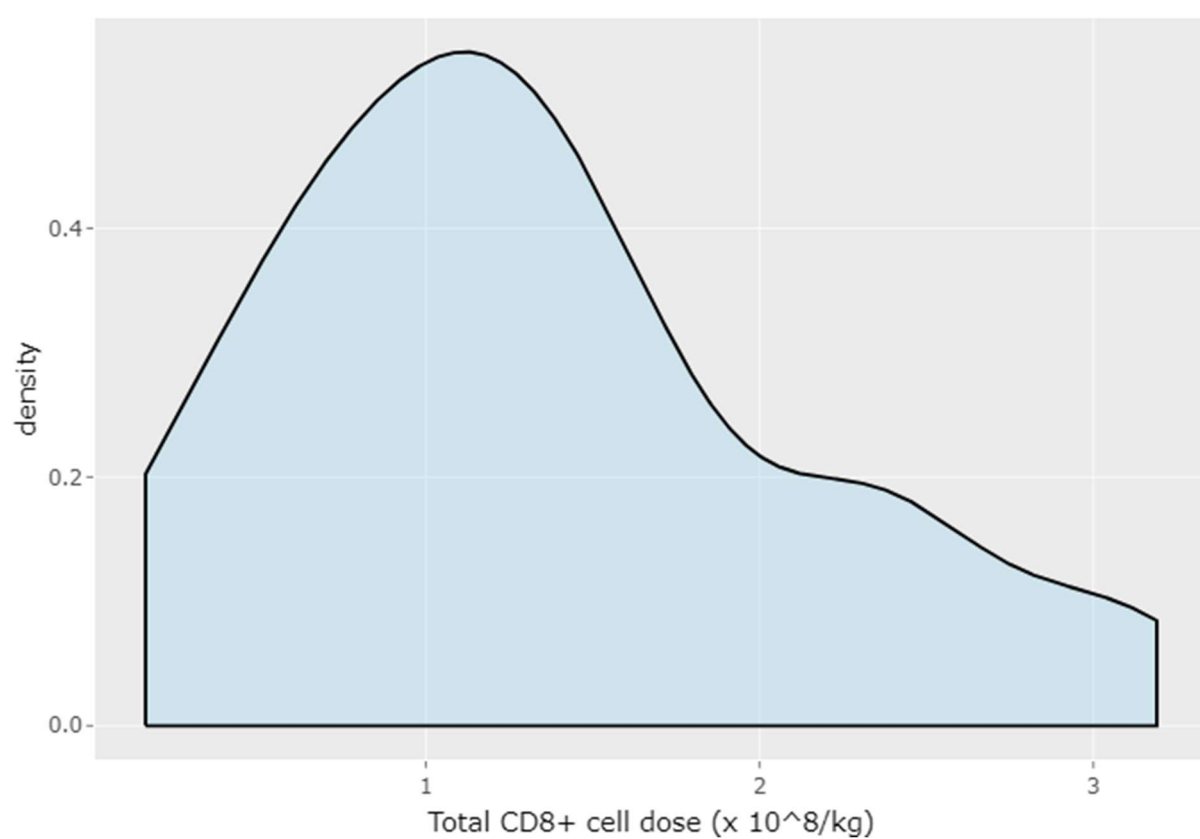
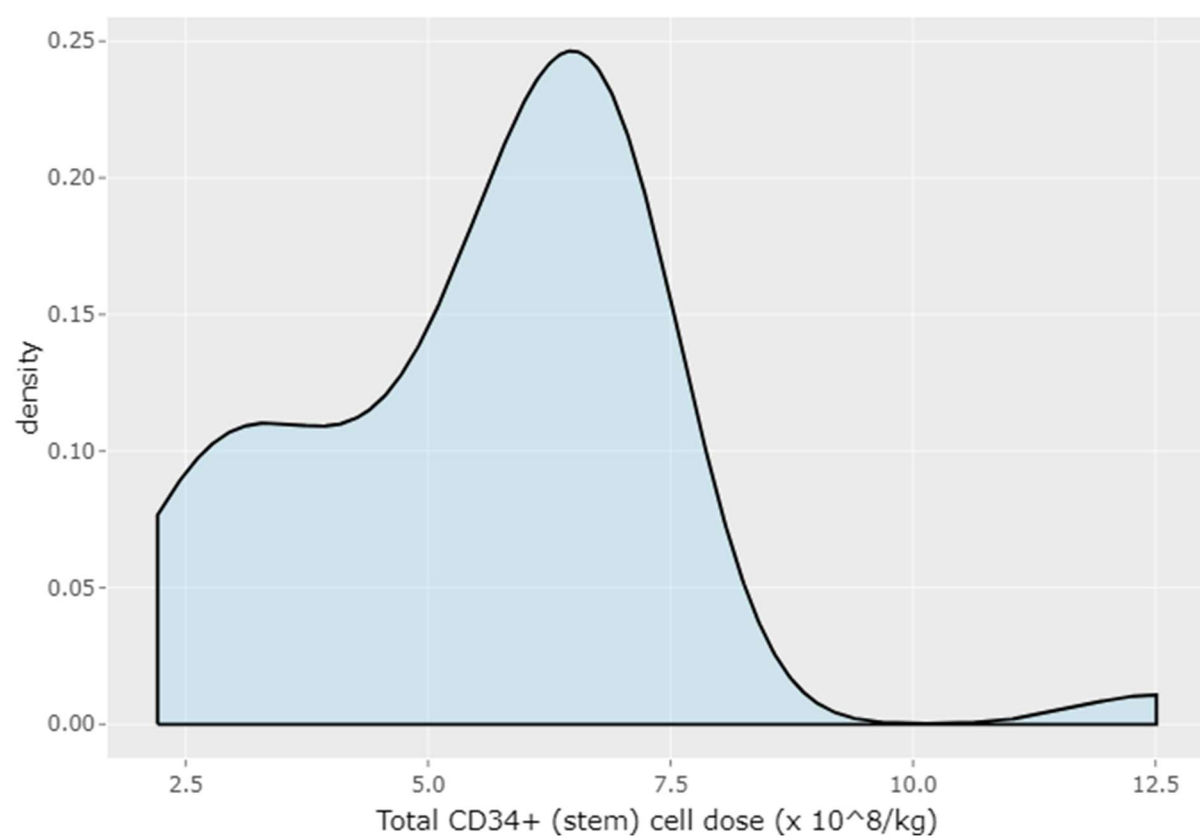
This correlation matrix shows that there is multicollinearity problem between :

TNC dose and CD8 dose,
CD3 dose and CD8 dose.

Density plots for continuous variables

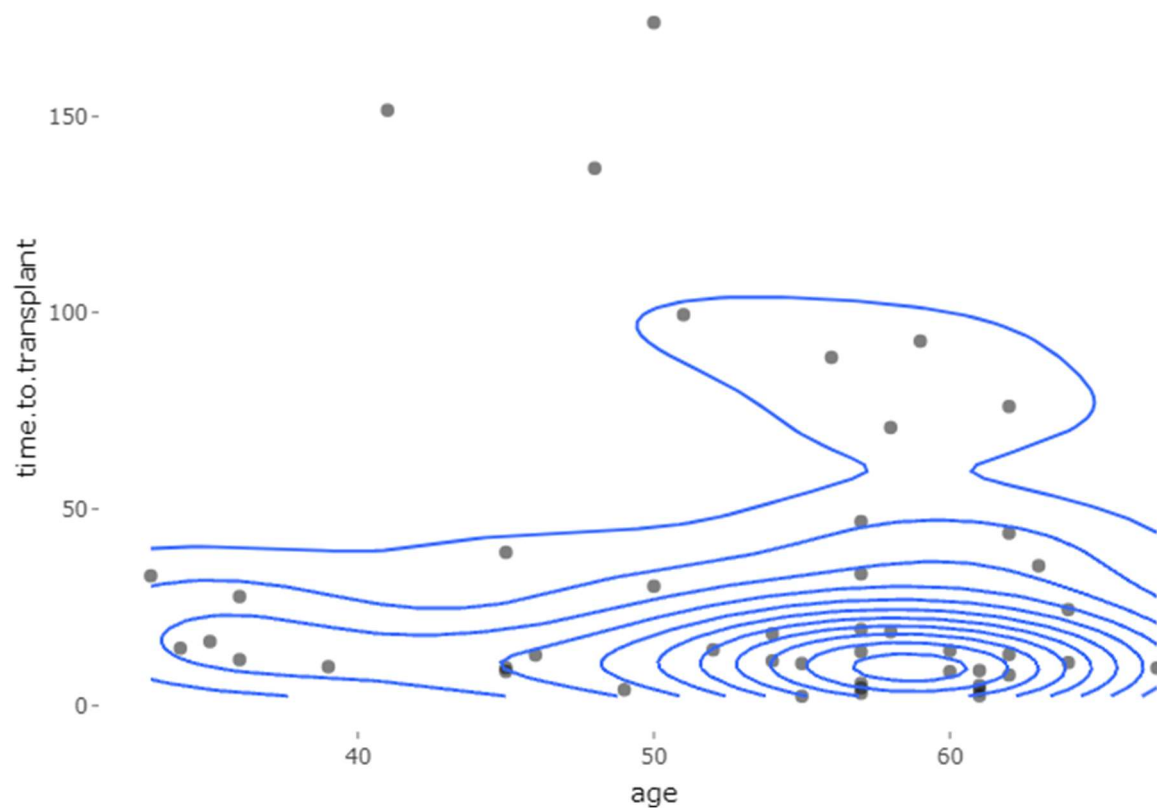






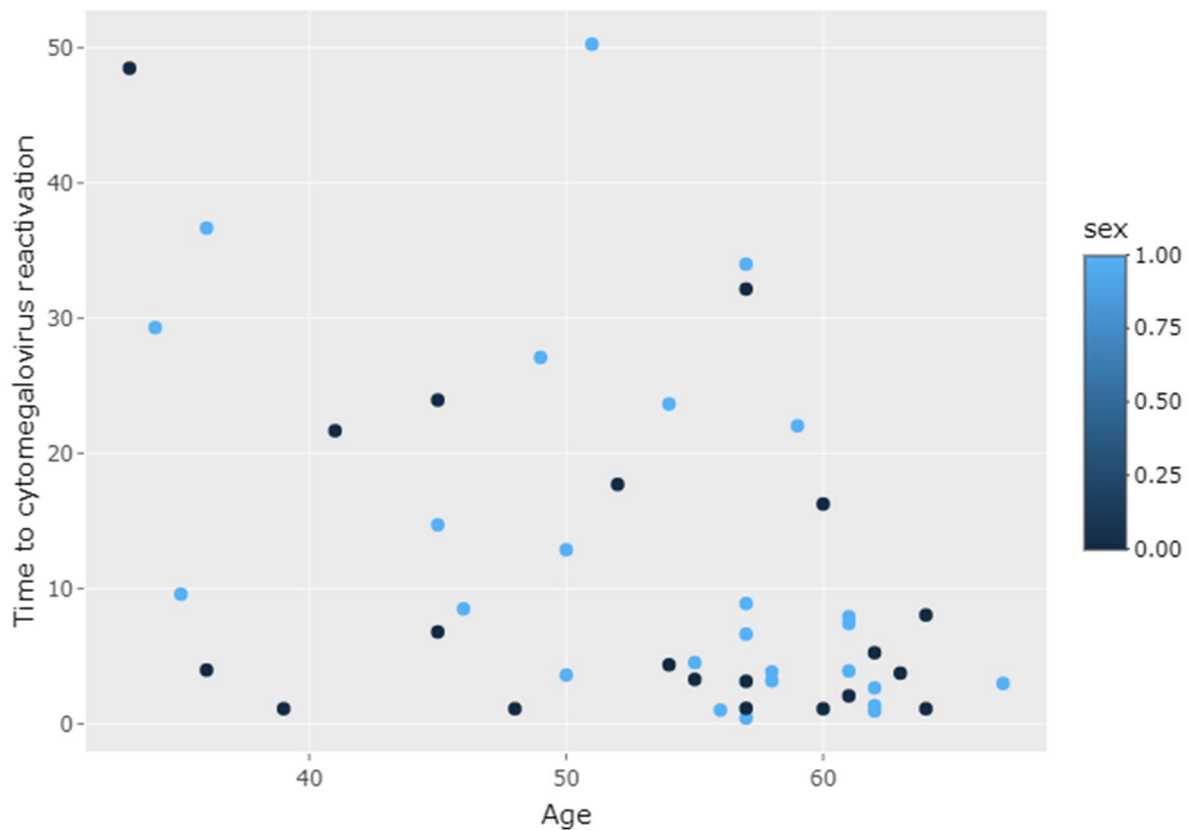
Some Relation plots

Age vs Time to transplant Plot.



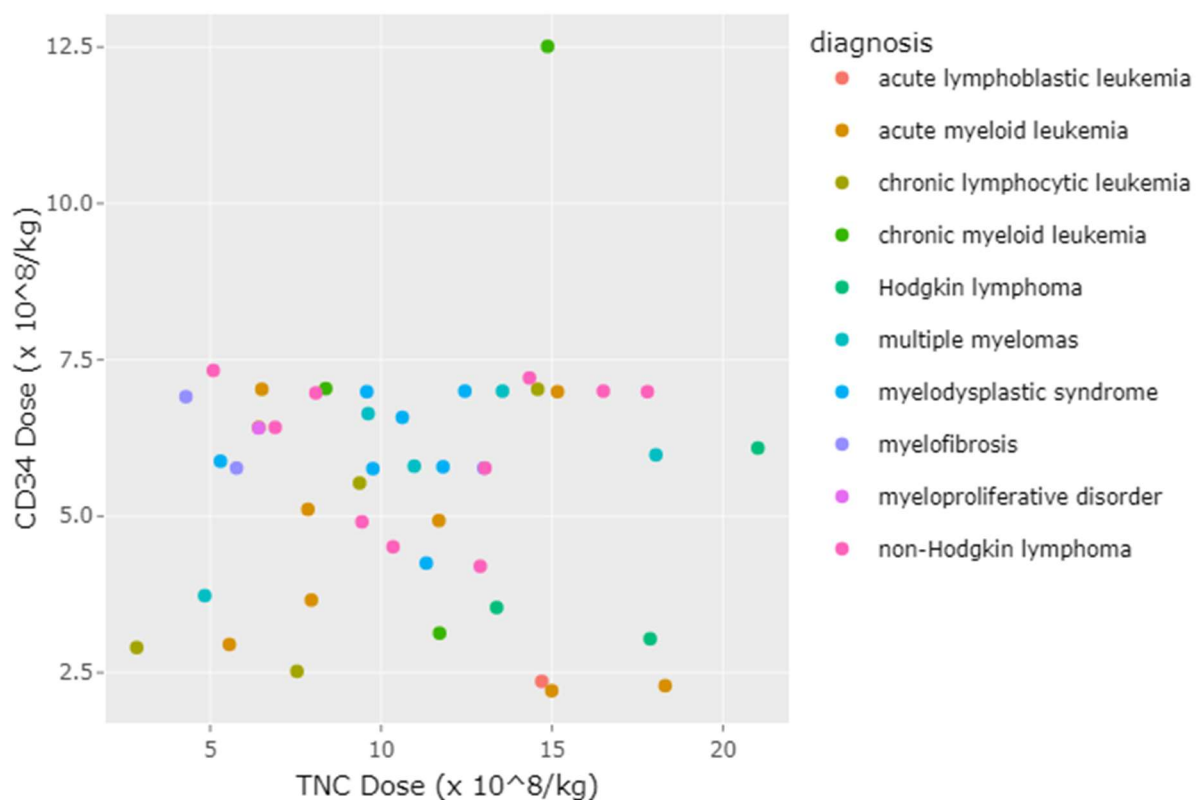
The density lines are concentrated at the age 60. Which means, if the age of patients are close to 60, the time from cancer diagnosis to transplant decreases. But still, there are some patients whose age are under 40 and their time from diagnosis to transplant is low.

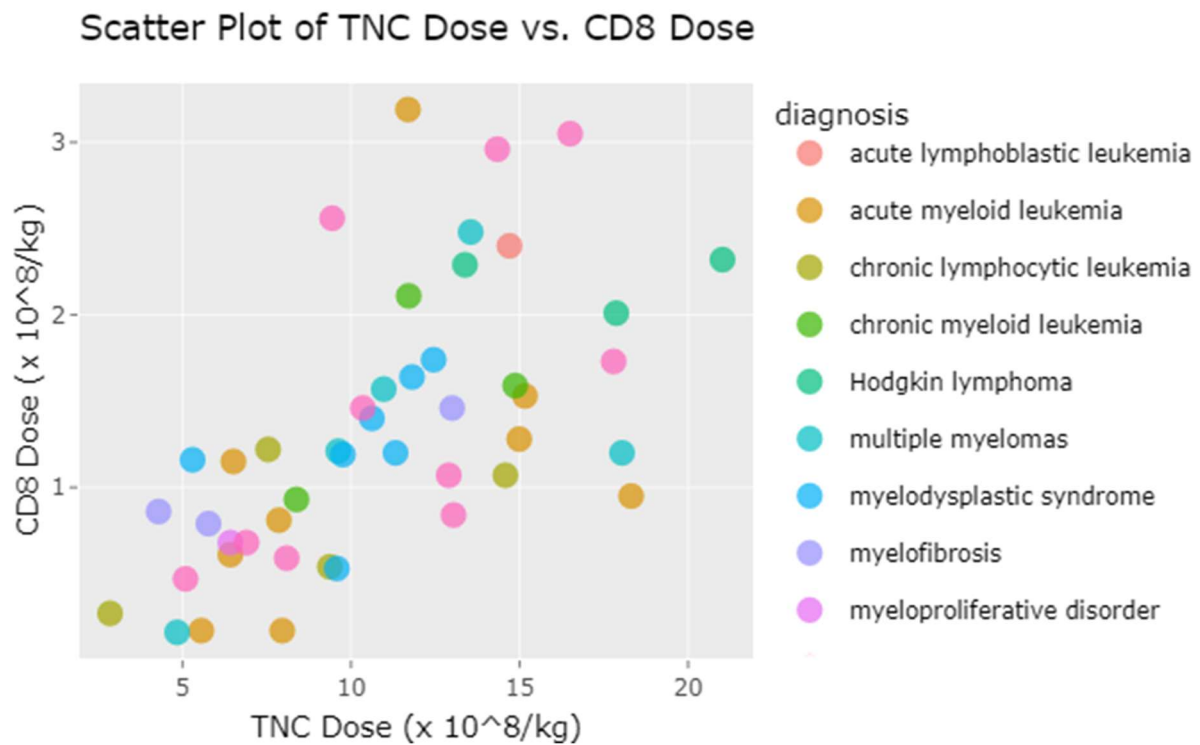
The plot for Age vs Time to Cytomegalovirus reactivation. 0 (male) , 1(female)



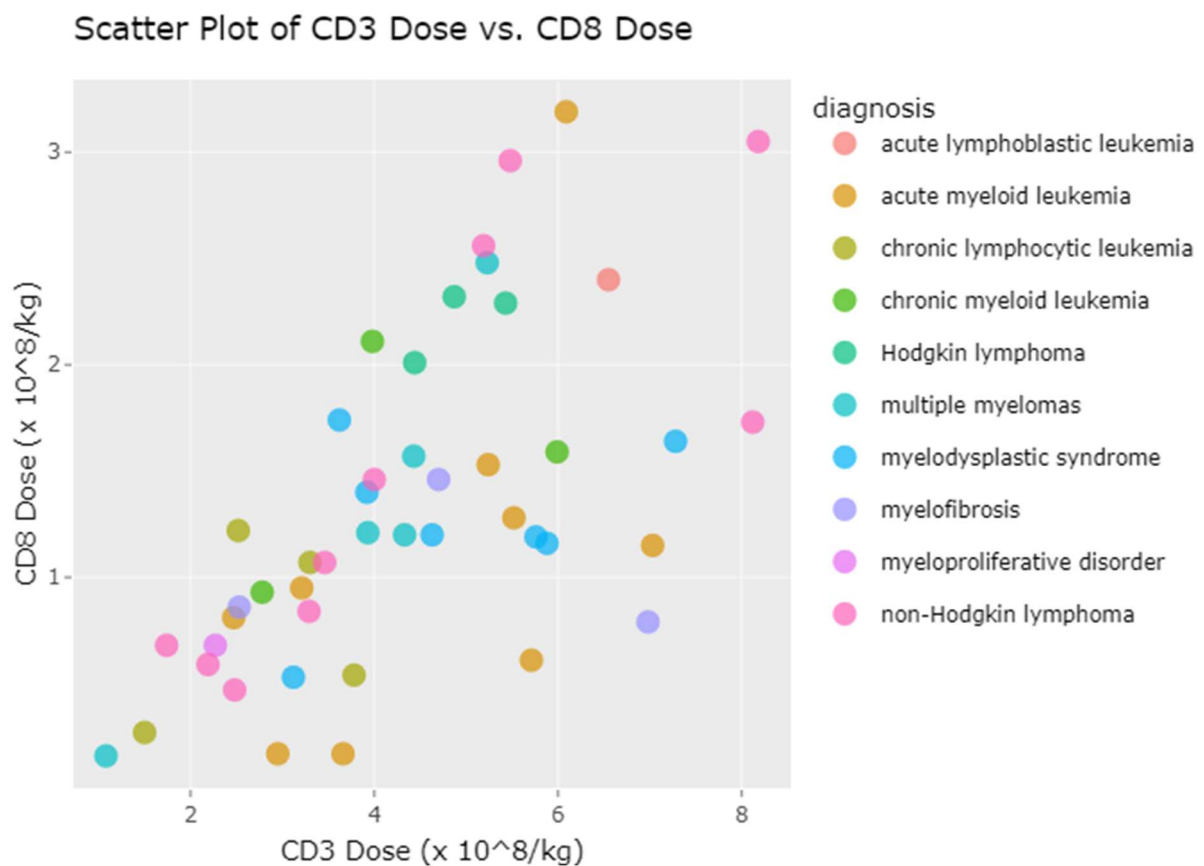
We can clearly see that there is no specific relationship between variables in both plots.

Scatter Plot of TNC Dose vs. CD34 Dose

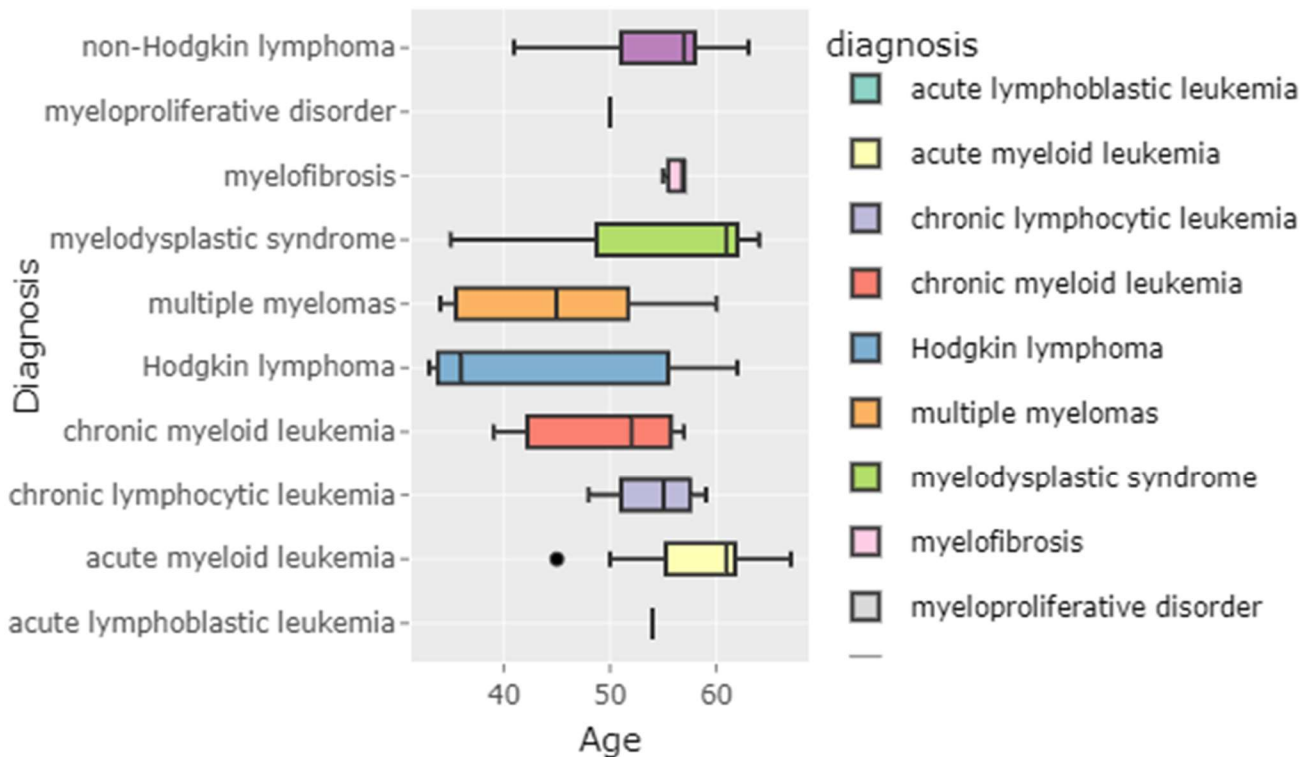




In these plots, we can clearly see that when TNC dose and CD3 dose increase, the CD8 dose also increases, which we observed in the correlation graph.

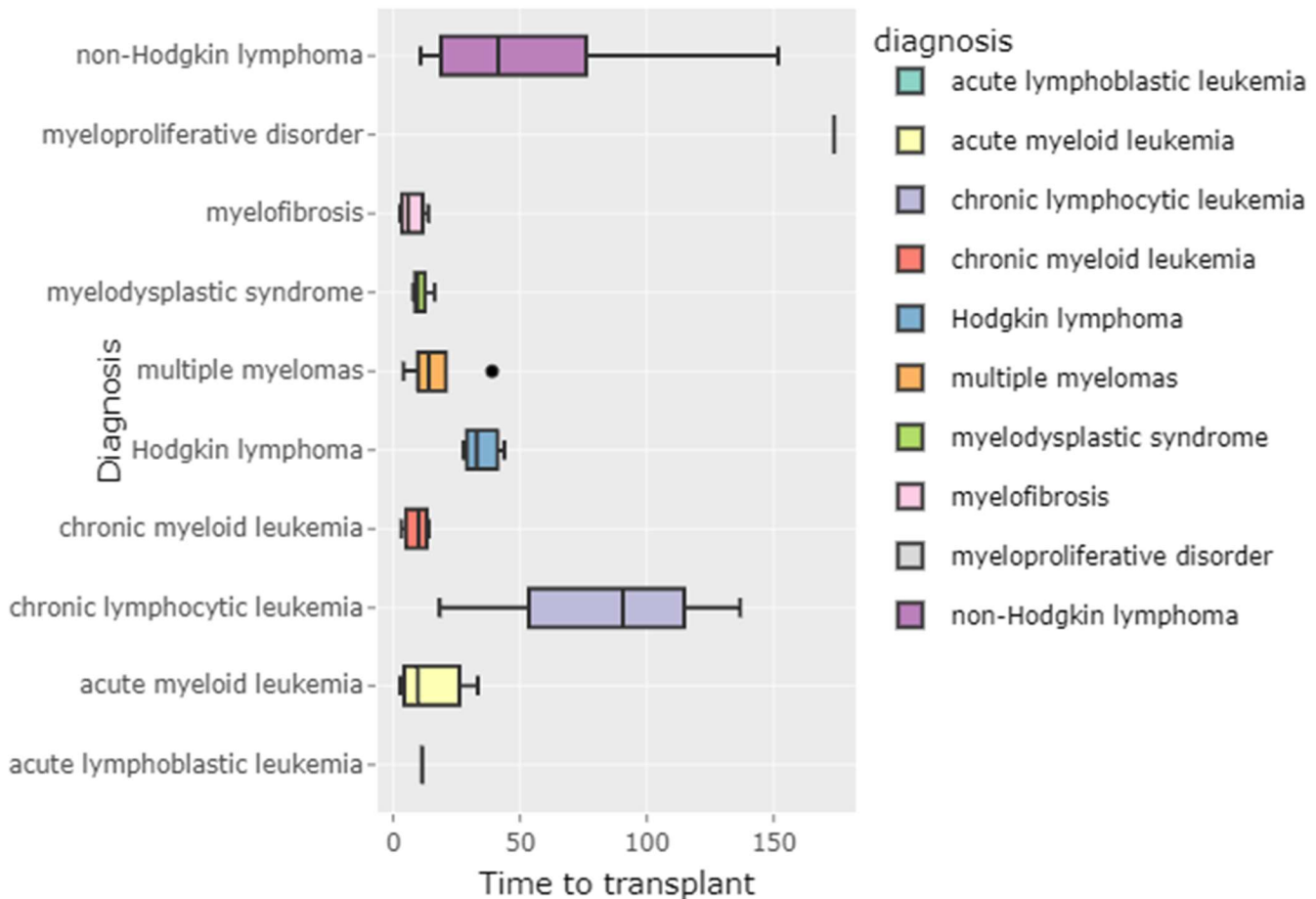


Age x Diagnosis Boxplot



The plot is indicating the age range varies from diagnosis to diagnosis. The graph might be misleading because the number of observations are not that much.

diagnosis x transplant time Boxplot



Contingency Table Analysis

We will take a look at whether diagnosis type depends on the gender or race (or not.)

First, we make a 2 by 2 table using gender and diagnosis type variables.

	Diagnosis Type		
		Myeloid	Lymphoid
Gender	Male	9	11
	Female	14	12

The odds ratio estimate : 0.7012987

The Wald Confidence Interval : (0.2175053 , 2.2611859)

The odds ratio indicates that females have approximately 70% of the odds of being diagnosed with Myeloid or Lymphoid type compared to males. However, 95% Wald Confidence Interval suggests that there is no significant association between gender and diagnosis type. Because the interval includes 1.

Now, let's test the independence of variables (gender and diagnosis type) using chi square test.

H0: The variables in row and column are independent

H0: The variables in row and column are NOT independent

The output :

```
      Pearson's Chi-squared test with Yates' continuity correction
data:  x2table
x-squared = 0.088462, df = 1, p-value = 0.7661
```

Our P-value is 0.76 ,which is greater than our alpha value 0.05 , we fail to reject null hypothesis. meaning that the variables are NOT associated.

Another a 2 by 2 table for race and diagnosis type variables.

	Diagnosis Type		
		Myeloid	Lymphoid
Race	White	3	1
	American/African	20	22

The odds ratio estimate : 3.3

The Wald Confidence Interval : (0.3169979 , 34.35354)

The odds ratio indicates that odds of American/African people being diagnosed with Myeloid or Lymphoid type is 3.3 times when compared to white people.

Again, since the Wald CI includes 1 as value in it , there is no significant association between race and diagnosis type.

Now, lets test the independence of variables (race and diagnosis type) using chi square test.

H0: The variables in row and column are independent

H0: The variables in row and column are NOT independent

Output:

```
Warning: Chi-squared approximation may be incorrect
```

We got a warning about Chi-square may be incorrect. Thats because Chi square statistic follows chi-squared distribution approximately. Chisq.test function warns us we have one expected count lower than 5. So, it is better to use Fisher's Exact Test.

```
output
Fisher's Exact Test for Count Data
```

```
data: x2table2
p-value = 0.6078
```

P value being greater than 0.05 indicates we fail to reject H_0 , which means the race and diagnosis type is also not related to each other.

Purposeful Model Building

Step 1) Constructing main effect models & collecting variables: (if $P < 0.2$)

I collected every variable using them as sole predictors. Constructed 25 models and got the p values of each variable. After collecting the p values, I saw that ID, age, sex, race, doses of TNC, DC3, DC8, DC34; time.to.transplant, prior.radiation, recipient.cmv, donor. sex, aKIRs, time.to.cmv variables are insignificant. (their P-values are > 0.2)

Fit1:

```
Call:
glm(formula = cmv ~ . - age - race - diagnosis - ID - sex -
time.to.transplant -
prior.radiation - recipient.cmv - donor.sex - TNC.dose -
CD3.dose - CD8.dose - CD34.dose - aKIRs - time.to.cmv, family =
"binomial",
data = traindata)
```

Coefficients:	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.647914	3.318878	-0.497	0.6195
diagnosis.type	-0.861635	1.836210	-0.469	0.6389
prior.chemo	-0.337678	0.482202	-0.700	0.4838
prior.transplant	-1.440730	1.513248	-0.952	0.3411
donor.cmv	1.668513	1.007119	1.657	0.0976 .
TBI.dose	-0.005605	0.005627	-0.996	0.3192
c1c2ratio	2.099070	1.171053	1.792	0.0731 .
agvhd	1.921550	1.448461	1.327	0.1846
time.to.agvhd	0.004604	0.095048	0.048	0.9614
cgvhd	-1.111302	1.448884	-0.767	0.4431
time.to.cgvhd	0.154186	0.106153	1.452	0.1464

Step2) Backward elimination:

I fitted a regression model using first fitted model with backward elimination method.

Fit2:

```
Call:
glm(formula = cmv ~ donor.cmv + c1c2ratio + agvhd + time.to.cgvhd,
family = "binomial", data = traindata)
```

Coefficients:	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-5.15232	1.59134	-3.238	0.0012 **

```

donor.cmv      1.91624    0.92472    2.072    0.0382 *
c1c2ratio     1.81777    0.93267    1.949    0.0513 .
agvhd         1.64441    0.95801    1.716    0.0861 .
time.to.cgvh  0.16769    0.07062    2.375    0.0176 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 54.846  on 40  degrees of freedom
Residual deviance: 34.744  on 36  degrees of freedom
AIC: 44.744

Number of Fisher Scoring iterations: 5
- time.to.cgvh  1    46.208 54.208

```

The method eliminated some variables and gave me a model shown in the output.

Step3) Adding variables that did not contribute in model 1

Fit3:

```

Call:
glm(formula = cmv ~ +age + race + ID + sex + time.to.transplant +
    prior.radiation + recipient.cmv + donor.sex + TNC.dose +
    CD3.dose + CD8.dose + CD34.dose + aKIRs - time.to.cmv + donor.cmv +
    c1c2ratio + agvhd + time.to.cgvh, family = "binomial", data =
    traindata)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.208e+02 3.577e+05 -0.001 1.000
age          -7.076e+00 1.387e+04 -0.001 1.000
race         1.654e+02 3.484e+05  0.000 1.000
ID           3.624e-01 3.965e+03  0.000 1.000
sex          1.603e+02 1.510e+05  0.001 0.999
time.to.transplant 9.831e-01 2.250e+03  0.000 1.000
prior.radiation -3.633e+02 4.876e+05 -0.001 0.999
recipient.cmv  2.783e+01 1.625e+05  0.000 1.000
donor.sex      2.642e+01 1.688e+05  0.000 1.000
TNC.dose       2.574e+00 3.545e+04  0.000 1.000
CD3.dose       5.607e+01 7.322e+04  0.001 0.999
CD8.dose      -9.527e+01 6.380e+04 -0.001 0.999
CD34.dose     -3.310e+01 3.394e+04 -0.001 0.999
aKIRs         -4.517e+01 8.875e+04 -0.001 1.000
donor.cmv      1.935e+02 1.112e+05  0.002 0.999
c1c2ratio      9.946e+01 1.729e+05  0.001 1.000
agvhd         2.009e+02 1.769e+05  0.001 0.999
time.to.cgvh   1.486e+01 9.054e+03  0.002 0.999

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 5.4846e+01  on 40  degrees of freedom
Residual deviance: 8.5078e-09  on 23  degrees of freedom
AIC: 36

Number of Fisher Scoring iterations: 25

```

Despite the AIC value, the model variables do not contribute to the model in any ways.

Now, lets construct a Likelihood Ratio Test (LRT) for decision.

Likelihood Ratio Test:

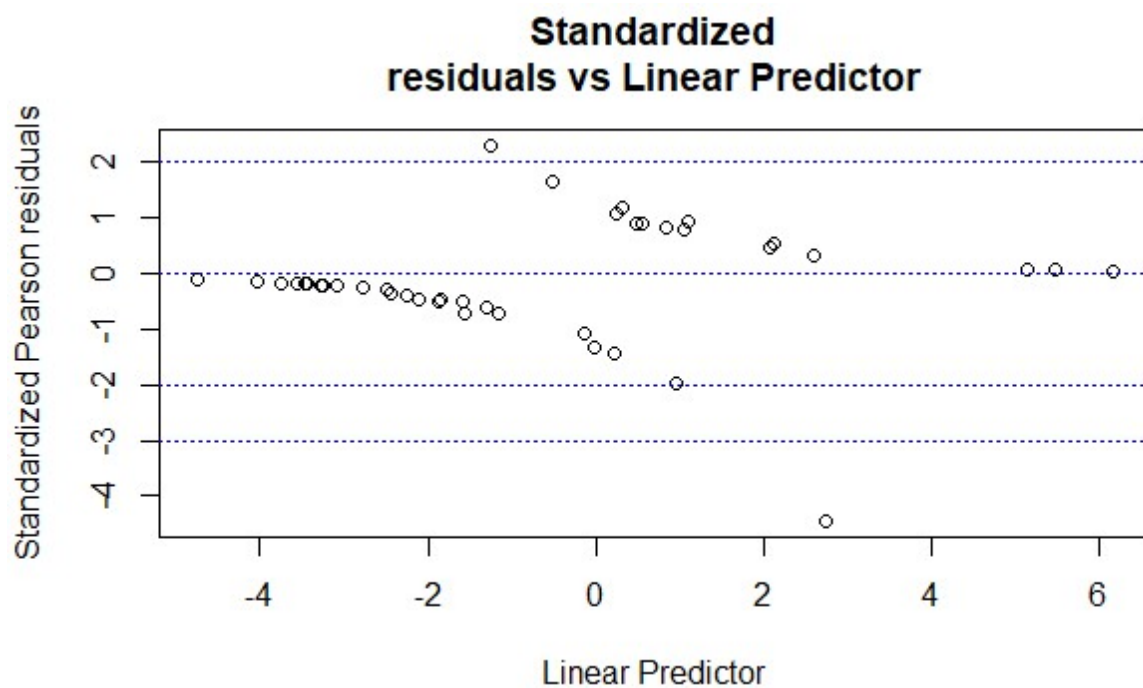
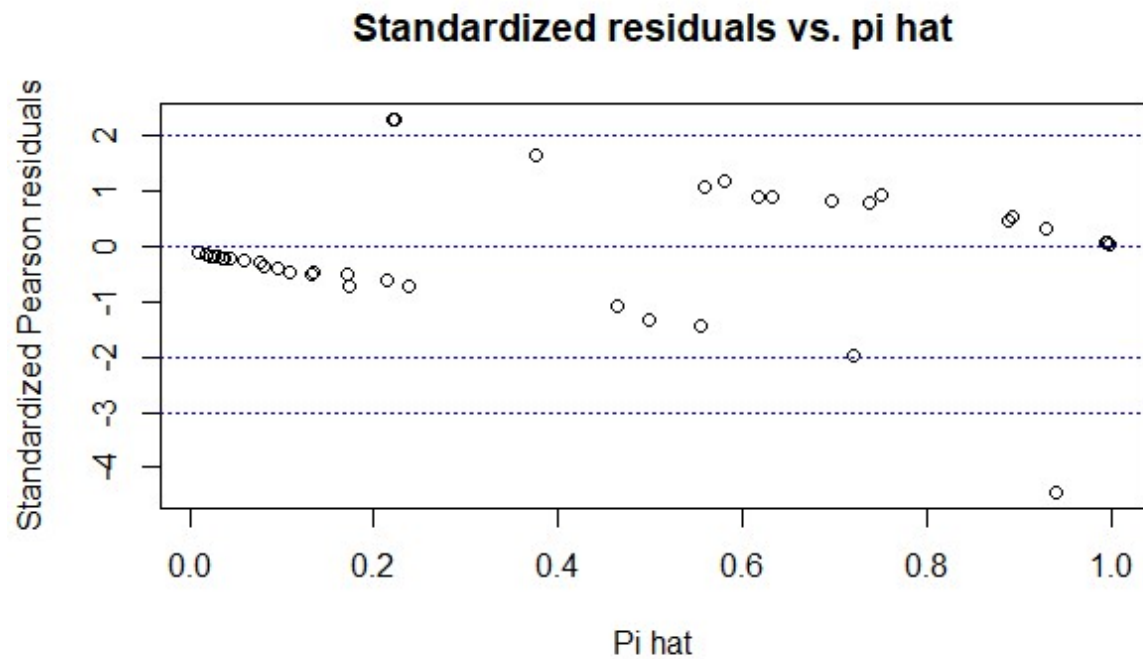
H0: full model is true.

H1: reduced model is true.

```
Likelihood ratio test
#Df  LogLik Df  Chisq Pr(>Chisq)
1   11 -15.416
2   18  0.000  7 30.833  6.675e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Since our P- value is (0.01264) < alpha value (0.05) we reject H0, Which means there significant difference in model fit between the full model and the reduced model. In other words, the reduced model (fit1) is better.

Diagnostics



Both plots show that there's only 2 outlier. Which might not be a big problem for the data & models.

R² value

Nagelkerke
0.6010176

Only 60% of the variation can be explained by our variables.

Obtainin predictions using Train & Test sets

The training set contains 41 observations and test set contains 5 observatitons

First 5 observations of train set. (first 5 columns are shown)

	ID <dbl>	age <dbl>	sex <dbl>	race <dbl>	diagnosis <chr>
1	1	61	1	0	acute myeloid leukemia
3	3	63	0	1	non-Hodgkin lymphoma
4	4	33	0	1	Hodgkin lymphoma
5	5	54	0	1	acute lymphoblastic leukemia
6	6	55	1	1	myelofibrosis

Predictions table

Cytomegalovirus\$Cmv table:

0	1
18	28

$$18/46 = 0.39$$

Meaning that for predictions, we need to choose threshold 0.4

Predictions table:

Pred_traindata	
0	1
23	18

Model Performances

	Accuracy	Sensivity	Specifity	F1 Score
Train	0.7560976	0.7600000	0.75	0.7916667
Test	0.8000000	0.6666667	1.00	0.8000000

The reduced model is seemed to have a good match for both the dataset and the unseen portion of the data because all train and test metrics are high enough.

4.Conclusion

In summary, I effectively created a logistic regression model in R to forecast the important predictor variables impacting CMV reactivation. These are also found and added to the model through extensive exploratory data analysis (EDA) and statistical testing. Based on patient demographics, the logistic regression model demonstrated encouraging performance metrics and offered meaningful information about the probability of CMV reactivation.

References

Peter Higgins [aut, cre] (). (2021, August 16). *Cytomegalovirus: Retrospective cohort study of the effects of donor kir... in medicaldata: Data Package for medical datasets.* cytomegalovirus: Retrospective Cohort Study of the Effects of Donor KIR... in medicaldata: Data Package for Medical Datasets.
<https://rdrr.io/cran/medicaldata/man/cytomegalovirus.html>

